

**VNUHCM, UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY**



AI APPLICATION REPORT

SUBJECT: INTRODUCTION TO INFORMATION TECHNOLOGY

INSTRUCTORS: Nguyễn Thanh Tình

Lê Nhựt Nam

CLASS: 25C01

GROUP: 1

No.	Student ID	Full name
1	25127200	Võ Nguyễn Nhất Huy (Leader)
2	25127113	Võ Thiện Trí Nhân
5	25127128	Nguyễn Nhật Quang
8	25127295	Nguyễn Lê Gia Bảo

Ho Chi Minh City - 12/2025

1 Group Information

Name: Hỉ Nộ Ái Ổ

Group: 1

STT	MSSV	Họ và Tên	Email
1	25127200	Võ Nguyễn Nhất Huy	vnnhuy2523@clc.fitus.edu.vn
2	25127113	Võ Thiện Trí Nhân	vttndan2518@clc.fitus.edu.vn
3	25127181	Nguyễn Lê Gia Bảo	nlgbao2523@clc.fitus.edu.vn
4	25127128	Nguyễn Nhật Quang	nnquang2538@clc.fitus.edu.vn

2 Project Information

2.1 Overall Plan

No.	Phase	Start – End Date	Goal	Description
1	General Planning & Concept Development	1/12/2025 – 2/12/2025	Decide the work that AI does; Select theme and overall visual direction; Decide platforms and API.	Team meeting to align ideas; Test different platforms (ChatGPT API, Google AI Studio) before deciding.
2	Production & Content Development	3/12/2025 – 6/12/2025	Complete API taking, develop simulation AI, deploy web; Complete slide structure/theme; Update project report.	Assign members to research/develop website; Group session for slides; Leader monitors progress and updates report.
3	Finalization & Review	6/12/2025 – 7/12/2025	Finalize website functions and test simulation; Finalize slide; Finalize written report.	Tasks assigned at start; Leader refines deliverables; Team reviews and evaluates project outcome.

2.2 Task Distribution and Completion Assessment

No.	Task	Date	Description	Assignee	Eval.
1	General Planning & Concept Development	1/12/2025 – 2/12/2025	Set overall plan; Decide AI product; Select theme, visual direction, and web functions.	Huy, Quang, Whole team	100%
2	Complete the AI interview simulation website	3/12/2025 – 6/12/2025	Optimize website: Design website, integrate API key, code visual layout.	Huy, Nhan, Bao	100%
			Finish slide: Design slide, prepare for presentation.	Bao, Quang, Nhan	
			Update report: Revise and add new information.	Bao, Nhan, Quang	
3	Finalize the website, presentation, and report	6/12/2025 – 7/12/2025	Complete all functions, test simulation; Finish slide contents; Complete final report. (Whole team must join).	Nhan (main), Quang & Bao (main), Huy (main)	100%

3 System Design and Architecture

Product: HCMUS AI Interviewer

3.1 System Overview

The platform used to build the AI: Google AI Studio

- At first, we prefer using OpenAI or Grok. However, they are required to buy tokens to interact with the user.

Programming Language: Python

- Python is a popular programming language used to develop Artificial Intelligence because of its advantages (various libraries, simple syntax, and there are a lot of coding communities that can help us with this project).

Platform to deploy the application: Streamlit

- Streamlit is a Python library that helps us to create interactive web apps without using "The Big Three" in web development (HTML, CSS, and JavaScript).
- The system follows a lightweight Client-Server architecture.

Large Language Models:

- Gemini 1.5 Pro (gemini-1.5-pro): Used for high-level reasoning tasks like generating interview questions and the final evaluation.
- Gemini 1.5 Flash (gemini-1.5-flash): Specifically used for OCR (reading text from images) in the CV upload section.
- Gemini 1.0 Pro (gemini-1.0-pro): Listed as a candidate for general generation tasks.

3.2 System Architecture Diagram

Our application's architecture is simply demonstrated in the diagram below

- The user interacts with the web app (1).
- After the web app receives the data, the data will be transferred to the AI. It will process and transfer the information back to the web app (2).

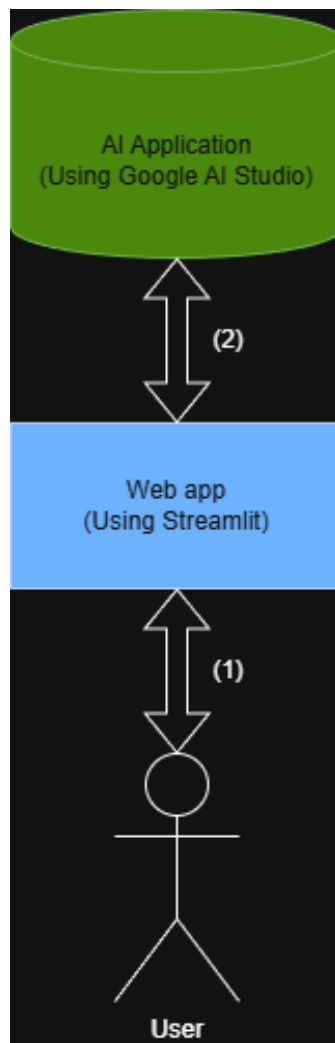


Figure 1: *Architecture Diagram*

3.3 Workflow

3.3.1 Information

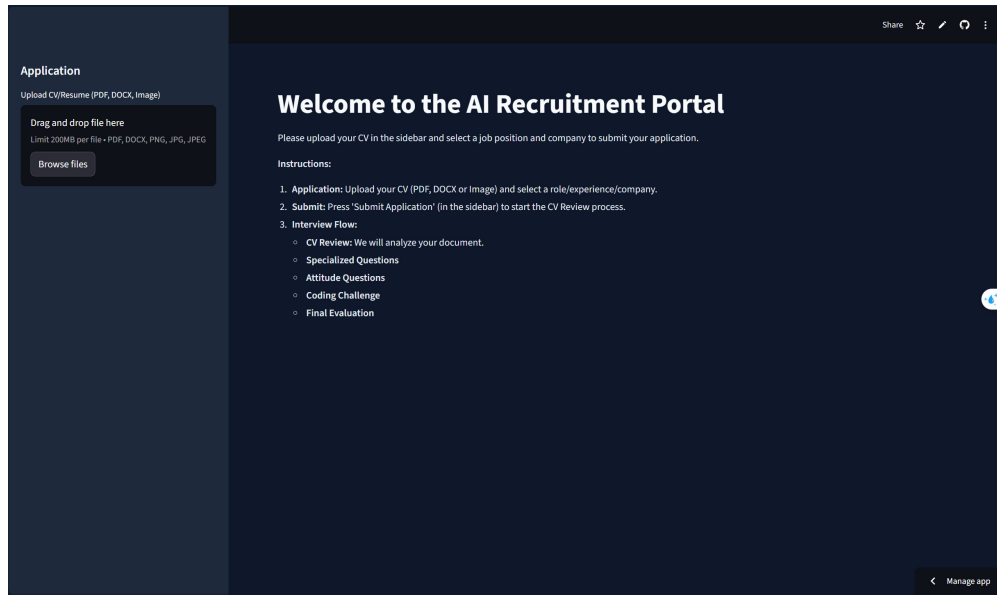


Figure 2: *Application Submission*

- The applicant sends his/her application (CV/Resume) to the web app through "*Browse files*" or by dragging and dropping the file onto the top left black box (Figure 2)

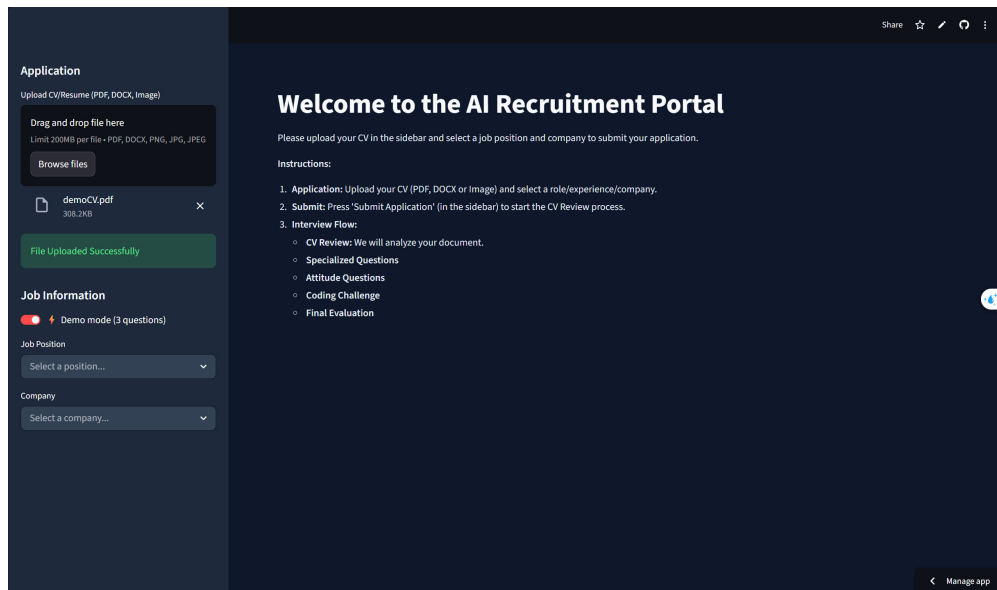


Figure 3: *Job Information*

- After the submission, the applicant must choose his/her job position and the company that he/she want to apply to. The questions will be given and evaluated based on the job position that the applicant has chosen (Figure 3).

- **There are 10 job positions:**

1. Artificial Intelligence Engineer
2. Backend Developer
3. Business Analyst
4. Cybersecurity
5. Data Scientist
6. DevOps Engineer
7. Frontend Developer
8. Fullstack Developer
9. Human Resource Manager
10. Software Developer

- **There are 8 companies:**

1. AMD
2. IBM
3. Intel
4. Meta
5. Microsoft
6. NVIDIA
7. VinAI Research
8. xAI

- There is a "Demo mode" that the applicant can choose to get used to the interface. The questions in this mode will be much easier, and the final evaluation won't be accepted.

3.3.2 Questions

After the applicant presses "Submit Application", the AI will take a while to process the application - 30 seconds if the "Demo Mode" is on, 3 minutes if the "Demo Mode" is off.

The reason why we choose 30 seconds and 3 minutes is because we want to replicate the actual environment of interviews. The interviewer needs some time to inspect through the application. In real life, the time may take longer. However, we are applying AI to the web app, so the time will be shortened. If the "Demo Mode" is on, the applicant just needs to get used to the interface. Therefore, the processing time can be shortened to 1 minute. After the inspection is complete, the applicant starts his/her interview.

There are three main parts:

1. **Specialized Questions**

10 questions if Demo Mode is off, 3 questions if Demo Mode is on.

2. Attitude Questions

30 questions if Demo Mode is off, 10 questions if Demo Mode is on.

3. Coding Challenge

3 questions if Demo Mode is off, 1 questions if Demo Mode is on.

There are two types of questions: Multiple choices and Text boxes.

3.3.3 Evaluations

After the applicant answers all the questions and submit, the AI will evaluate the answers after 3 minutes (if Demo Mode is off) and 1 minutes (if Demo Mode is one). The scores will be showed to the applicant (Application score, Specialize Questions score, Attitude Questions score and Coding Challenge score) besides the decision of the company (which is either HIRED or NOT HIRED). In order to be hired, the overall score must be higher than 7.0. To calculate the overall score, the AI will calculate the average of 4 scores.

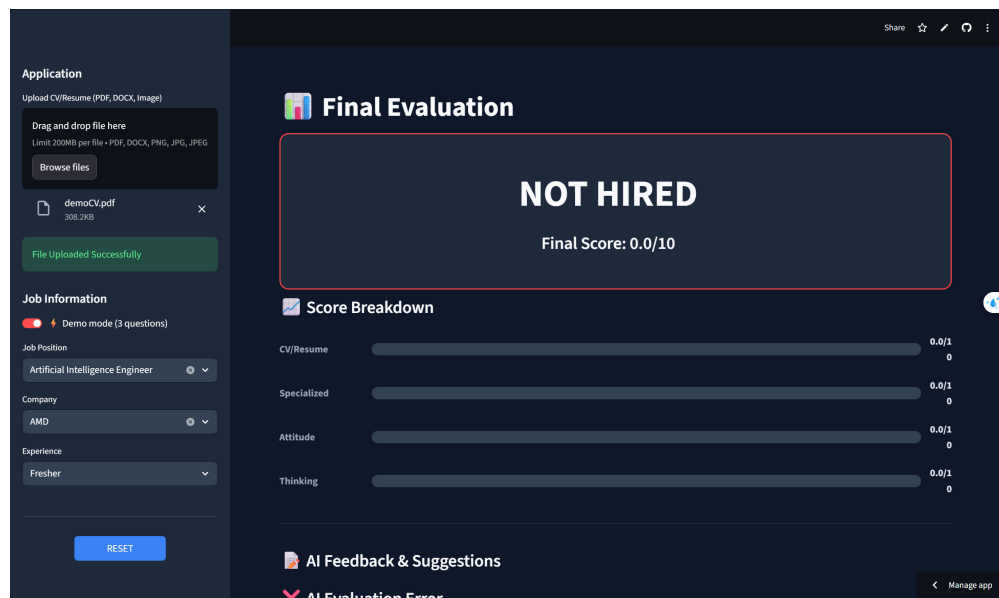


Figure 4: Final Evaluation

4 Dataset and Models

For this project, the product utilizes the Google AI Studio API, which provides access to Google's pretrained large language models.

These models are trained on a large-scale, diverse dataset consisting of publicly available text, licensed data, and data created by human trainers, enabling the model to understand natural language and generate contextually appropriate responses.

Instead of building and training a custom model or dataset, the project leverages a pretrained language model via API to simulate interview scenarios. The model is responsible for generating interview questions, responding to user answers, and maintaining a coherent and realistic conversational flow.

This approach allows the system to focus on interaction design and user experience while relying on Google's robust pretrained model for natural language understanding and generation.

Using Google AI Studio helps ensure reliable performance, scalability, and ease of integration, making it suitable for developing an AI-powered interview simulation website without the need for complex model training or dataset preparation.